

Q41 - Clique Partitioning Problem (NP-Hard Problem)

Aim: To partition graph vertices into cliques using backtracking.

Constraints: Every vertex belongs to exactly one clique and all clique members are mutually connected.

Approach: Explore clique assignments using backtracking.

Pruning: Add a vertex only if it connects to every vertex in the clique.

Complexity: Exponential in worst case.

NP-Hard: Optimal clique partitioning requires exploring many combinations.

Conclusion: Backtracking with pruning efficiently eliminates infeasible partitions.

Output Screenshot: Insert screenshot here.